PARV DIXIT

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Education

B.Tech, Electrical Engineering, IIT Jodhpur	**CGPA: 8.28/10**	2024 – Present
CBSE Class 12, Shriji Baba Saraswati Vidya Mandir	**Percentage: 94.6%**	2024
CBSE Class 10, Shriji Baba Saraswati Vidya Mandir	**Percentage: 99.2%**	2022

Experience

Autonomous Warehouse Inspection Robot [GitHub] — Oct 2025 – Dec 2025
*Inter IIT Tech Meet 14.0 | Problem Statement by Eternal | **6th Rank** among 23 IITs*

- Engineered a 2WD differential-drive AMR achieving **+/-7 cm horizontal** and **+/-1 cm vertical localization accuracy** in GPS-denied indoor environments.
- Architected ROS 1 (Jetson Nano) -> `ros1_bridge` -> ROS 2 Nav2 (NavFn + DWB, 20 Hz) 3-tier stack for real-time autonomous navigation.
- Fabricated a **belt-driven T-slot gantry** (up to **2 m**, **16 s**/sweep) with Arduino stepper control, hardware E-Stop, and HMI via Jetson Nano.

LiDAR SLAM and Autonomous Navigation Robot [GitHub] — Jan 2026 – Apr 2026
Research under Dr. Riby Abraham Boby, Dept. of Mechanical Engineering, IIT Jodhpur

- Deployed **ROS 2 Humble** AMR with **SLAM Toolbox** for 2D occupancy-grid mapping at 5 cm resolution via RP LiDAR A1 (360 deg, 8000 sps).
- Integrated **Nav2** (A* + DWA, 20 Hz) for autonomous navigation, obstacle avoidance, and goal-directed path execution.
- Implemented **ESP32 micro-ROS node** for 4x quadrature encoding (1440 CPR), 50 Hz odometry, and Modbus-UART motor control at 115200 baud.

Mentor – Robotic Arm Painter [GitHub] — May 2026 – Present
Robotics Society Summer Project, IIT Jodhpur | 15+ students, 6-week curriculum

- Built CV pipeline (CLAHE -> Bilateral -> Canny -> contour resampling) for stroke trajectory extraction on a **3-DoF robotic arm**.
- Devised phi-sweep **IK** with B-spline/Chaikin smoothing and 3-layer velocity clamp (<=**100 deg/s**) for safe stroke execution.
- Commissioned **ESP32** serial (115200 baud, ACK-gated) driving 3+1 servo axes with arc-length stepping (0.015 AU); mentored **15+ students** over 6 weeks.

Projects

Flight Controller Enhancement – Pluto Nano Drone [GitHub] — Nov 2025 – Dec 2025
Inter IIT Tech Meet 14.0 | Problem Statement by Drona Aviation | Cleanflight Firmware

- Constructed a dual-loop state estimator on Cleanflight (**250 Hz**) fusing IMU, optical flow, ToF, and barometer with a **3rd-order complementary filter** for drift-free altitude and bias correction.
- Incorporated **ZUPT** and quality-weighted optical-flow/IMU blending; cascaded **Position -> Velocity -> Attitude PID** with **DCM**-based estimation for GPS-denied hover.

Autonomous Power Grid Control – Deep Reinforcement Learning [GitHub] — Mar 2026 – Apr 2026
Soft Actor-Critic | C++ Physics Simulator | ZeroMQ | PyTorch

- Trained a **SAC agent** (Twin Critic + auto-tuned entropy, tau=0.005, 26K-step warm-up) managing a **100-substation grid** over 8,760-step episodes via **ZeroMQ** REQ/REP IPC.
- Crafted a **7-component shaped reward** (demand service, overload penalty, cascade prevention, utilisation health) clamped to [-10,10] with checkpoint/resume support.

A.E.O.L.U.S – Custom Quadrotor Flight Controller [GitHub] — Aug 2025 – Oct 2025
MATLAB Simulink | Model-Based Design | Embedded Control | Robotics Society, IIT Jodhpur

- Modelled nonlinear quadrotor plant in Simulink with tilt-compensated thrust.
- Designed cascaded attitude–rate **PID** with complementary filter (alpha=0.9995) fusing gyro, accelerometer, and magnetometer for drift-free attitude estimation.

GTK Chess Engine & Power Grid Visualizer [GitHub] — Aug 2024 – Nov 2024
C | GTK3 | Cairo | C++ | Raylib | Data Structures & Algorithms

- Authored a **GUI chess engine** in C with GTK3/Cairo, legal-move validation, check/checkmate, and logic-driven AI.
- Developed a **C++/Raylib power-grid simulator** with widest-path overload rerouting and LOD rendering.

Technical Skills

Languages	C, C++, Python, Verilog
Frameworks/Libraries	PyTorch, OpenCV, ZeroMQ, Gymnasium, GTK3, Raylib
Tools	ROS 2 (Nav2, SLAM, micro-ROS), Gazebo, RViz2, MATLAB Simulink, Vivado, Git
Embedded	ESP32, Jetson Nano, UART/Modbus, Encoder Interfacing
Control	PID, Sensor Fusion (IMU, LiDAR, ToF, Optical Flow), State Estimation

Positions of Responsibility

Assistant Head, *Prometeo '26 – Technical Fest, IIT Jodhpur* — *Dec 2025 – Jan 2026*
Spearheaded technical event operations managing 10+ volunteers across multiple concurrent events.

Assistant Head, *Sandstone '25 – Annual Business Conclave, IIT Jodhpur* — *Aug 2025 – Sep 2025*
Coordinated accommodation, transport, and security verticals for guest speakers at the annual business conclave.